Midterm review problem set

Student handout

From Euclid to finite fields

16 review problems | 160 practice points + 20 optional | Suggested time: 5–7 hours over several sittings

Preparation. Earlier lectures and assignments, ending with finite-fields Lecture 2: independence, bases, dimension, prime-power cardinalities, and quadratic extensions.

Name: _____ **Due:** _____

How to use this set. This is a review bank, not a timed mock exam. Try each problem before reading a solution; explain every use of a theorem. Write on separate paper. The point values indicate relative weight. You may quote results proved in class except when explicitly asked to prove them. The final two optional problems revisit the earlier enrichment set. Finite-field cyclicity, matrix counts, Frobenius, and Fibonacci period arguments from Lectures 3–4 are outside this set's scope.

Notation. $v_p(m)$ is the exponent of the prime p in a nonzero integer m; extend it to nonzero rationals by $v_p(a/b) = v_p(a) - v_p(b)$. The functions φ, τ, σ count units, count positive divisors, and sum positive divisors, respectively. The Möbius function satisfies $\mu(1) = 1$, vanishes on integers divisible by a prime square, and equals $(-1)^r$ on a product of r distinct primes.

I. Integers and continued fractions

1. Euclid gives a certificate [10 points]

(a) Compute $\gcd(252, 165)$ by Euclid and express it as an integer linear combination of 252 and 165. (5)

(b) Find all solutions of $165x \equiv 6 \pmod{252}$, giving the distinct residues modulo 252. Explain why there are exactly that many. (5)

2. What factorization proves [10 points]

(a) Prove that $\sqrt{72}$ is irrational by comparing prime exponents in a hypothetical equality $a^2 = 72b^2$, with a, b nonzero integers. (4)

(b) Prove there are infinitely many primes congruent to 3 modulo 4. *Hint:* given a proposed complete list $p_1, \ldots, p_r$, consider $N = 4p_1 \cdots p_r - 1$. Explain why some prime divisor of N is 3 modulo 4; do not assume N is prime. (6)

3. A finite continued fraction remembers Bézout [10 points]

(a) Find the simple continued fraction of $157/68$ and all of its convergents. Use the convention that the last digit is at least 2. (4)

(b) Multiply the digit matrices $D(a) = \left(\begin{smallmatrix} a & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Use the determinant of their product to find the inverse of 68 modulo 157. (4)

(c) Compute exactly $157/68 - 30/13$. (2)

4. Periodic is not always purely periodic [10 points]

For $x_n = (P_n + \sqrt{D})/Q_n$ and $a_n = \lfloor x_n \rfloor$, recall

$$P_{n+1} = a_n Q_n - P_n, \qquad Q_{n+1} = (D - P_{n+1}^2)/Q_n.$$

(a) Run this recurrence for $\sqrt{11}$ until a state repeats, and give its continued fraction. Justify the floors with integer bounds. (5)

(b) Decide which of $\sqrt{11}$, $3+\sqrt{11}$, and $(3+\sqrt{11})/2$ have purely periodic expansions. Use Galois's criterion $x > 1$ and $-1 < x' < 0$, where x' is the conjugate. Read off the purely periodic expansions from (a). (5)

II. Congruences, powers, and arithmetic functions

5. Reassemble residues, and check cancellation [10 points]

(a) Solve $x \equiv 3 \pmod 8$, $x \equiv 4 \pmod 9$, and justify uniqueness modulo 72. (5)

(b) Find every solution of $x^2 \equiv 1 \pmod{15}$ using reduction modulo 3 and 5. Why does the argument "$(x-1)(x+1) = 0$, so one factor is zero" fail modulo 15? (5)

6. Fermat and inverse pairing [10 points]

(a) Compute $7^{222} \pmod{13}$ efficiently. (3)

(b) Prove Wilson's theorem $(p-1)! \equiv -1 \pmod p$ for a prime p by pairing inverses. Account for $p = 2$. (5)

(c) The composite number 341 satisfies $2^{340} \equiv 1 \pmod{341}$. Explain why this is consistent with Fermat's little theorem. (2)

7. Orders explain exponent reduction [10 points]

(a) Find the orders of 2 modulo 5, modulo 7, and modulo 35. Justify the relation between the three orders. (4)

(b) Find $\varphi(35)$ and compute $2^{1000} \pmod{35}$. (3)

(c) Explain how Lagrange's theorem applied to $(\mathbb{Z}/n\mathbb{Z})^\times$ proves Euler's theorem. State the hypothesis on the base. (3)

8. Work one prime at a time [10 points]

(a) Factor 360 and compute $\tau(360), \sigma(360), \varphi(360)$, $\mu(360)$, and $\mu(30)$. Show the prime-power calculations. (5)

(b) Prove that if f is multiplicative, then $F(n) = \sum_{d \mid n} f(d)$ is multiplicative. Explain precisely where coprimality is used. (5)

9. Invert divisor sums and count exact periods [10 points]

(a) Suppose $\sum_{d\mid n} f(d) = n^2$. Use Möbius inversion to give a formula for $f(n)$, and evaluate $f(12)$. (4)

(b) Let $A(n)$ count labeled binary words of length n with least rotational period exactly n. Explain why $2^n = \sum_{d\mid n} A(d)$. Find $A(6)$ and the number of binary necklaces of length 6 with least period 6, where rotations are identified. Explain the final division. (6)

III. Valuations, primes, and p-adic distance

10. Count prime factors without expanding factorials [10 points]

(a) State Legendre's formula. Use it to find $v_2(40!)$ and the number of trailing decimal zeros of $40!$. (3)

(b) Put $B_n = \binom{2n}{n}$. Prove that each summand in

$$v_p(B_n) = \sum_{j\ge 1} \left(\left\lfloor \frac{2n}{p^j} \right\rfloor - 2 \left\lfloor \frac{n}{p^j} \right\rfloor \right)$$

is 0 or 1, and deduce $B_n \mid \operatorname{lcm}(1, \dots, 2n)$. (4)

(c) For prime p, prove $p \mid \binom{p}{k}$ when $0 < k < p$. Deduce $(X+Y)^p = X^p + Y^p$ as a polynomial identity in characteristic p. (3)

11. Recover the scale of the primes [10 points]

Let $\pi(x) = \#\{p \le x : p \text{ prime}\}$ and $\vartheta(x) = \sum_{p\le x} \log p$, with natural logarithms.

(a) Explain why the central binomial coefficient is the largest in its row, and prove $4^n/(2n+1) \le B_n < 4^n$ for $n \ge 1$. (3)

(b) Use Problem 10 to prove (4)

$$\pi(2n) \ge \frac{2n\log 2 - \log(2n+1)}{\log(2n)}.$$

(c) You may use the lecture estimate $\vartheta(x) \le Ax$ for a fixed constant $A > 0$ and all $x \ge 2$. Show $\pi(x) \le \sqrt{x} + 2Ax/\log x$. Explain why these estimates give $\pi(x) \asymp x/\log x$ but do not give $\pi(x) \sim x/\log x$. (3)

12. Close in a different arithmetic [10 points]

For nonzero rational r, define $|r|_3 = 3^{-v_3(r)}$; set $|0|_3 = 0$ and $d_3(x,y) = |x-y|_3$.

(a) Compute $|27/20|_3$ and $d_3(1/4, 7/4)$. (2)

(b) Prove $|x+y|_3 \le \max(|x|_3, |y|_3)$ for rational x, y. Prove equality if $|x|_3 \ne |y|_3$. (4)

(c) Find the 3-adic limit in $\mathbb{Q}$ of $S_n = \sum_{j=0}^{n} 3^j$ by computing the distance to your proposed limit. Does S_n have a finite real limit? (4)

IV. Finite fields: through Lecture 2

13. Units, zero divisors, and characteristic [10 points]

(a) List the units of $\mathbb{Z}/12\mathbb{Z}$ and exhibit two nonzero residues whose product is zero. (2)

(b) Prove that a finite field has prime characteristic. Explain how its multiples of 1 form a copy of $\mathbb{F}_p$. (4)

(c) A finite commutative ring R has $1 \neq 0$ and no nonzero zero divisors. Prove it is a field by considering multiplication by a fixed nonzero element. Where is finiteness used? (4)

14. Why dimension is well defined [10 points]

(a) In $\mathbb{F}_5^2$, show that $u = (1, 2)$ and $v = (2, 1)$ form a basis. Find the coordinates of $(3, 4)$ in this basis. (3)

(b) Let $v_1, \ldots, v_r$ be independent and let $w_1, \ldots, w_m$ span a vector space over a field. Reproduce the exchange argument proving $r \leq m$: after inserting the first k independent vectors, why can the next insertion remove an old w rather than a previously inserted v? Deduce that any two finite bases have the same length. (7)

15. Count coordinates, not residue classes [10 points]

(a) Prove every finite field K has p^d elements for some prime p and integer $d \geq 1$. Explain why a basis over $\mathbb{F}_p$ exists and why counting its coordinates counts each element exactly once. (6)

(b) If a field has 81 elements, determine its characteristic and its dimension over its prime field. Can a field have 12 elements? Does the prime-power argument by itself construct a field of every prime-power size? (4)

16. Build a field with twenty-five elements [10 points]

Use formal pairs $a+bs$ with $a, b \in \mathbb{F}_5$, and the relation $s^2 = 2$. Addition is coefficientwise and multiplication is distributive with s^2 replaced by 2. You may use the resulting commutative ring laws.

(a) List the squares in $\mathbb{F}_5$. Prove that every nonzero $a + bs$ has inverse $(a - bs)/(a^2 - 2b^2)$, explaining why the denominator is nonzero. (4)

(b) Compute $(1 + s)(2 + 3s)$ and $(1 + s)^{-1}$, and check the inverse by multiplication. (2)

(c) Give a basis over $\mathbb{F}_5$, the cardinality, and the characteristic of this field. Explain why it differs from $\mathbb{Z}/25\mathbb{Z}$. (2)

(d) If we instead imposed $t^2 = 4$ on formal pairs over $\mathbb{F}_5$, would the resulting ring be a field? Give explicit evidence. (2)

Optional review of earlier enrichment

These two problems review the assignment explicitly labeled optional enrichment; they do not introduce a later finite-fields lecture.

Optional challenge: Inclusion–exclusion as inversion [10 extra points]

In a group of 40 people, the numbers in activities A, B, C are $20, 18, 15$; the pairwise intersection counts are $9, 7, 6$, and the triple intersection count is 3. Pairwise counts include the triple intersection. Find the number in exactly one activity, exactly two activities, and no activities. Explain the alternating signs as recovering exact counts from counts that include additional memberships.

Optional challenge: Interpolation and a finite Fourier transform [10 extra points]

In $\mathbb{F}_5$, let $\omega = 2$, whose powers are $1, 2, 4, 3$. A polynomial f of degree at most 3 satisfies $f(1) = 1$, $f(2) = 0$, $f(4) = 0$, $f(3) = 0$. Recover f using either Lagrange interpolation or

$$c_k = 4^{-1} \sum_{j=0}^{3} f(\omega^j)\omega^{-jk}, \qquad f(X) = \sum_{k=0}^{3} c_k X^k.$$

Check all four values. Why is the degree restriction necessary for uniqueness? Give another polynomial of higher degree with the same values.